

SmartDRM: Blockchain based Digital Rights Management System

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Abstract

- ▶ Rapid growth of digital content distribution has intensified piracy, ownership violations, and revenue leakage
- ▶ Conventional Digital Rights Management (DRM) systems rely on centralized authorities for control and enforcement
- ▶ Centralized DRM architectures suffer from single-point failure, limited transparency, and reduced stakeholder trust
- ▶ SmartDRM-X introduces a decentralized DRM framework to address these structural limitations
- ▶ Smart contracts provide automated, tamper-resistant license enforcement mechanisms
- ▶ An Artificial Intelligence layer analyzes usage patterns to detect suspicious or abnormal behavior
- ▶ Cryptographic techniques ensure data integrity and protection against unauthorized modification

Introduction

- ▶ Explosive growth of digital media platforms has reshaped content creation, distribution, and consumption
- ▶ Unauthorized copying, illegal redistribution, and misuse of digital assets continue to rise
- ▶ Traditional DRM systems rely on centralized control, reducing transparency and system resilience
- ▶ Content creators lack visibility and users must blindly trust a single authority
- ▶ Advances in blockchain and AI enable SmartDRM-X: a decentralized, intelligent, and transparent DRM framework

PROBLEM STATEMENT :

- Centralized control leading to single-point failure
- Lack of transparent ownership verification
- Manual and unreliable license enforcement
- No intelligent mechanism to detect abnormal usage patterns
- Limited auditability and accountability

OBJECTIVE :

- ▶ To design a decentralized DRM framework using blockchain technology
- ▶ To ensure immutable ownership and license records through smart contracts
- ▶ To provide controlled and flexible access mechanisms for digital assets
- ▶ To detect suspicious usage behavior using AI-based analysis
- ▶ To maintain an auditable and transparent activity log
- ▶ To develop a modern, user-friendly interface for asset management

Existing System

- ▶ Current DRM solutions primarily operate using centralized servers managed by content distributors or platform providers.
- ▶ Characteristics:
- ▶ Central authority controls access and licenses
- ▶ User authentication and permissions stored centrally
- ▶ DRM policies enforced at application level

Disadvantage

- ▶ Single-point failure and high vulnerability to attacks
- ▶ Limited transparency in ownership verification
- ▶ Dependency on trusted third parties
- ▶ Difficult to audit access and usage
- ▶ Poor scalability for large digital ecosystems

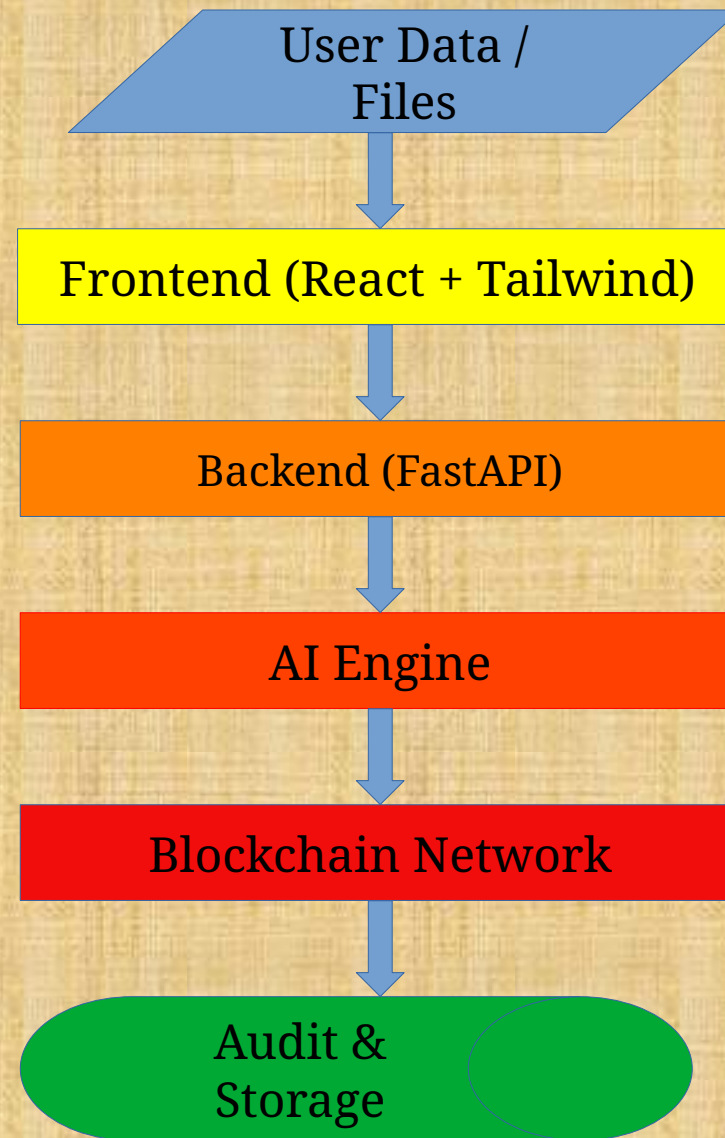
PROPOSED SYSTEM :

- ▶ Blockchain-based asset registration and ownership proof
- ▶ Smart contract-driven license issuance and enforcement
- ▶ AI-powered behavioral risk analysis
- ▶ Secure audit logging for all critical actions
- ▶ Full-stack web interface for real-time interaction

Advantage

- ▶ Immutable and tamper-proof ownership records
- ▶ Transparent license enforcement
- ▶ Reduced piracy and unauthorized access
- ▶ Automated royalty-ready transaction tracking

SYSTEM ARCHITECTURE /FLOW DIAGRAM :



MODULES :

- ▶ Authentication Module
- ▶ Asset Management Module
- ▶ Blockchain & Smart Contract Module
- ▶ License & Access Control Module
- ▶ AI-Based Risk Analysis Module
- ▶ Audit Logging Module
- ▶ Frontend Dashboard Module

MODULE 1:Authentication Module

- ▶ Purpose: Secure onboarding and identity verification of all users accessing protected digital assets.
- ▶ Key Features: Email/password login, secure password hashing, and session/token-based authentication.
- ▶ Role-Based Access: Differentiates admins, content owners, and consumers with tailored permissions.
- ▶ Security Controls: Brute-force protection, validation of credentials, and secure storage of user data.
- ▶ Integration Points: Acts as the entry gate to all other modules (assets, blockchain, AI risk, and dashboard).

MODULE 2: Asset Management

Module

- ▶ Purpose: Central hub to register, encrypt, store, and manage digital content.
- ▶ Asset Lifecycle: Upload, metadata registration, encryption, storage, and controlled distribution.
- ▶ Content Protection: Uses cryptographic techniques to prevent unauthorized duplication and sharing.
- ▶ Search & Organization: Metadata-based cataloging for quick discovery and filtering of assets.
- ▶ Integration Points: Links assets to licenses, blockchain records, and audit logs.

MODULE 3: Blockchain & Smart Contract Module

- ▶ Purpose: Provide tamper-evident, verifiable records for licenses and asset transactions.
- ▶ Smart Contracts: Encode licensing terms (duration, usage limits, regions) as executable logic.
- ▶ Immutable Ledger: Stores key events like license issuance, revocation, and critical asset operations.
- ▶ Transparency & Trust: Enables independent verification of rights and reduces disputes.
- ▶ Integration Points: Consumed by license/access control and audit modules for verification.

CONCLUSION

- ▶ SmartDRM-X demonstrates how blockchain and artificial intelligence can be effectively integrated to build a secure, transparent, and intelligent Digital Rights Management system. By decentralizing ownership records, automating license enforcement, and introducing behavioral analysis, the system significantly improves trust and security in digital content distribution.
- ▶ The proposed framework is scalable, extensible, and suitable for applications in education, digital media platforms, enterprise document management, and software licensing. Future enhancements may include integration with decentralized storage systems and advanced machine learning models for predictive piracy detection.



THANK YOU !